

CHRIS WILKERSON

Technical Writer and Illustrator

Colorado Springs, CO | wilky9@gmail.com | 719-401-7796 | linkedin.com/in/chriswilkerson-9982a23

01 - USER MANUALS

Power Meter Battery Replacement

DUB-PWR Battery Replacement
Wechsel der DUB-PWR-Batterie
Sustitución de la batería del DUB-PWR

Remplacement de la pile du DUB-PWR
Sostituzione della batteria DUB-PWR
De DUB-PWR-batterij vervangen

Substituição da bateria do DUB-PWR
DUB-PWR 電池の交換
DUB-PWR 電池更換



1. Use fingers, or a spanner wrench if necessary, to turn the battery sled cap of the battery sled clockwise 90 degrees, until the dot on the battery sled cap is aligned with the recessed dot on the battery sled.
2. Pull the battery sled out of the non-drive side crank spindle by hand.
3. Remove the AAA Lithium battery.
4. Install a new battery in the correct orientation.

NOTICE
Do not allow water to enter the spindle while the battery sled is loose or removed.

AVIS
Dès que le chariot de la pile est desserré ou retiré, veillez à ce qu'aucune trace d'humidité ne pénètre dans l'axe.

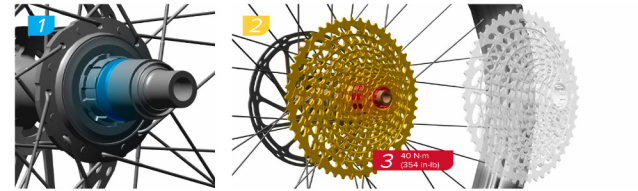
NOTIFICAÇÃO
Não deixe que entre água na cavilha-eixo enquanto o invólucro deslizante para a bateria estiver solto ou retirado.

XD/XDR Cassette Installation Procedure

Cassette Installation
Einbau der Kassette
Instalación del casete

Installation de la cassette
Installazione della cassetta
De cassette installeren

Instalação da Cassete
カセットの取り付け
飞轮安装



SRAM XD® Cassettes: Apply grease to the driver body threads.
Install the cassette onto the driver body until it is fully seated with the splines engaged, then tighten the cassette.

Cassettes SRAM XD®: appliquez de la graisse sur le filetage du corps de cassette.
Installez la cassette sur le corps de cassette en veillant à bien engager les cannelures puis serrez la cassette.

Cassetes SRAM XD®: Aplique massa lubrificante nos fios de rosca do corpo propulsor/accionador.
Instale a cassete no veio propulsor até que fique completamente assente com as estrias engatadas, e depois aperte a cassete.

NOTICE
Grease is critical to prevent the cassette from seizing onto the driver body. Make sure the hub driver body and cassette interface are free of dirt and debris.

AVIS
Il est très important de lubrifier la cassette pour éviter qu'elle se bloque dans le corps de cassette. Veillez à ce que le corps de cassette du moyeu et l'interface de la cassette ne présentent ni poussière ni débris.

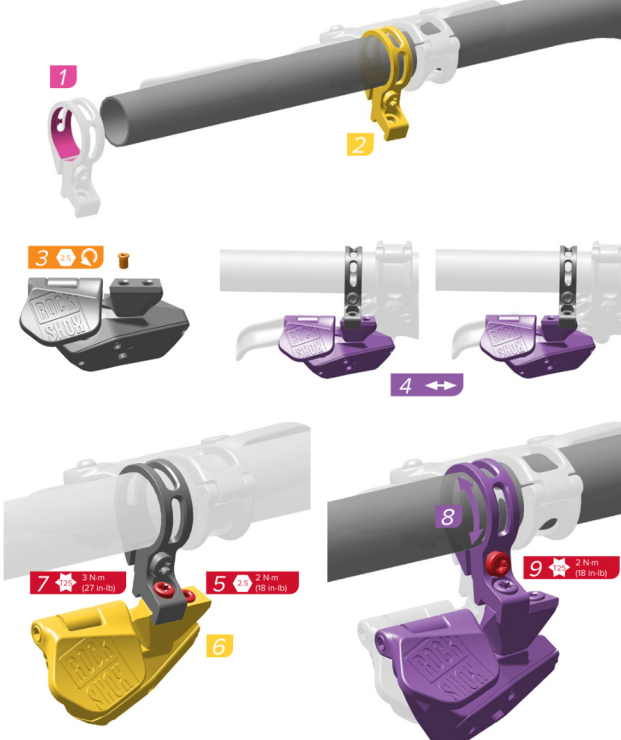
NOTIFICAÇÃO
A massa lubrificante é crítica para evitar que a cassete gripe (se agarre) no corpo do accionador. Assegure-se de que o corpo da manga propulsora do cubo e a interface da cassete estejam livres de sujidade e de detritos.

Wireless Shifter Installation Procedure

AXS Controller - Discrete Clamp
Separate Klemmschelle für AXS-Controller
Abrazadera Discreta del controlador AXS

Commande AXS - Collier Discrete
Controller AXS - Morsetto Discreto
AXS-controller - Discrete klem

Controlador AXS - Braçadeira Discreta
AXS コントローラー・ディスクリート・クランプ
AXS 致动器 - 分立式夹具



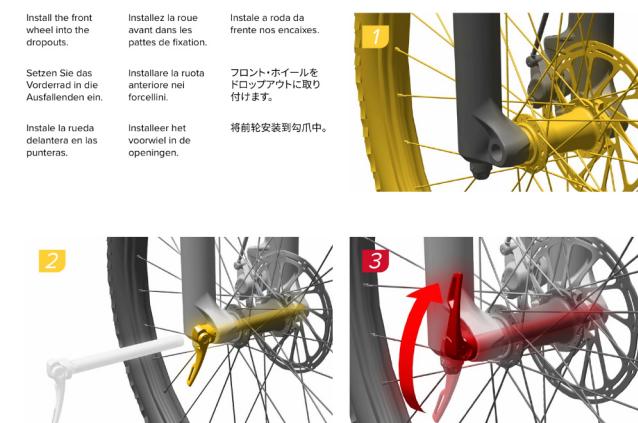
1. Attach the discrete clamp to the handlebar.
2. Attach the AXS controller to the handlebar.
3. Attach the AXS shifter to the handlebar.
4. Attach the AXS shifter to the handlebar.
5. Attach the AXS shifter to the handlebar.
6. Attach the AXS shifter to the handlebar.
7. Attach the AXS shifter to the handlebar.
8. Attach the AXS shifter to the handlebar.
9. Attach the AXS shifter to the handlebar.

Front Wheel Installation Procedure

Front Wheel Installation
Einbau des Vorderrads
Instalación de la rueda delantera

Installation de la roue avant
Installazione della ruota anteriore
Het voorwiel installeren

Instalação da roda frente
フロント・ホイールの取り付け
前轮安装

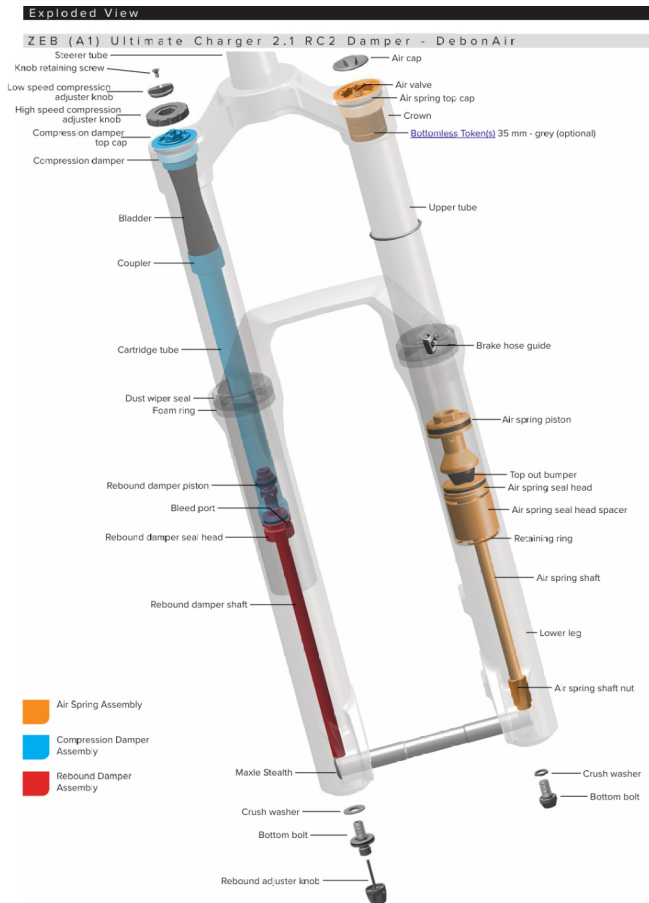


Install the front wheel into the dropouts.
Setzen Sie das Vorderrad in die Ausfallenden ein.
Instale la rueda delantera en las punteras.

Installez la roue avant dans les pattes de fixation.
Installare la ruota anteriore nei forcellini.
Installeer het voorwiel in de openingen.

Instale a roda da frente nos encaixes.
フロント・ホイールをドロップアウトに取り付けます。
將前轮安装到勾爪中。

ZEB (A1) Suspension Fork Internal View with Labeled Parts



Zipp Rear Hub Partial Bearing Installation Procedure

Rear Bearing Installation

NOTICE

To prevent damage when pressing the bearings into the rear hub, make sure that the bearing press tool contacts both the inner and outer races of the bearing.

- 1** Install a new Zipp 6803 bearing into the non-drive side of the hub with the **black** seal facing outward.
- Note:** Ceramic bearings have **blue** seals on both sides of the bearing; installation orientation is not important.



- 2** Install a 6903 30x17 tool into the drive-side bearing bore. Insert the threaded rod through the drive side of the hub shell. Thread the Press Tool handle onto the threaded rod. Turn the handle clockwise to press the bearing into the hub until it is hand-tight. Do not overtighten the bearing. Remove the tools.



NOTICE

To prevent damage when pressing the bearing into the hub, make sure that the bearing press tools contact both the inner and outer bearing races or bearing bores and not the hub shell.



ZEB (A1) Suspension Fork Partial Air Spring Service Procedure

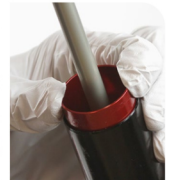
- 7** Apply grease to the air piston and seal head outer o-rings/seals. Apply suspension oil to the spring shaft, above the seal head.

CAUTION

Always wear safety glasses. Do not allow oil, fluid, grease, lubricant, or cleaner to contact your eyes or face. Seek immediate medical attention if irritation occurs.



- 8** Insert the air spring assembly into the upper tube. Firmly push the air piston into the upper tube. Insert the red spacer into the upper tube and firmly press it into the upper tube until it stops.

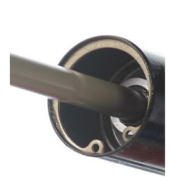


- 9** Retaining rings have a sharper-edged side and a rounder edged side. Installing retaining rings with the sharper-edged side facing the tool will allow for easier installation and removal.

Place the tips of the retaining ring pliers into the eyelets of the retaining ring. Guide the retaining ring with your finger to prevent the shaft from getting scratched while installing the retaining ring.

Use the pliers to push the seal head into the upper tube while installing the retaining ring into the groove. Release the retaining ring pliers when the ring is fully seated in the groove.

Confirm the retaining ring is properly seated in the retaining ring groove by using the retaining ring pliers to rotate the retaining ring and seal head back and forth a few times, then firmly pull down on the air shaft.



NOTICE

Do not scratch the air spring shaft. Scratches on the air shaft will allow air to bypass the seal head into the lower leg, resulting in reduced spring performance.

WARNING - CRASH HAZARD

Retaining rings must be fully seated in the retaining ring groove. Confirm the retaining ring is fully seated in the retaining ring groove after installation. Failure to do so can result in **SERIOUS INJURY OR DEATH**.

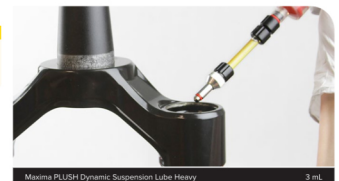
- 10** Pull the shaft out until it stops.



- 11** Inject or pour suspension oil into the air spring upper tube.

CAUTION

Always wear safety glasses. Do not allow oil, fluid, grease, lubricant, or cleaner to contact your eyes or face. Seek immediate medical attention if irritation occurs.



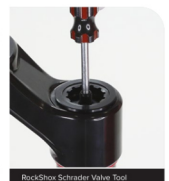
- 12** Install the air spring top cap into the upper tube and tighten it. Press down firmly when tightening the top cap.

WARNING - CRASH HAZARD

Parts must be tightened to the specified torque. Failure to do so can result in **SERIOUS INJURY OR DEATH**.



- 13** Install the Schrader valve core into the top cap and tighten it finger tight.



03 - ARCHITECTURAL VISUALIZATION

Chinook Trail Elementary - CSNA Architects



Chinook Trail Elementary - Finished Building Photograph

04 - LINKS TO DOCUMENTS

Source documents, hosted on SRAM and CSNA Architects websites, for the spreads shown in this portfolio.

[Eagle Transmission - User Manual](#)

[RockShox ZEB - Service Manual \(2021–2022\)](#)

[Eagle AXS Systems - User Manual](#)

[Road AXS and XPLR AXS - User Manual](#)

[Zipp MTB Wheels - User Manual](#)

[Mineral Oil MTB Disc Brake Installation, Hose Shortening & Bleed Manual](#)

[Chinook Trail Elementary School](#)

05 - USER MANUAL DESIGN PHILOSOPHY

1. **User Comfort and Delight:** Based on surveys and focus groups that repeatedly cited a dislike for text-heavy instructions, I placed an emphasis on visual elements over written instruction. To reduce the amount of text, a standard set of colors was applied to all user manual illustrations, with each color representing a specific action the user will perform.
2. **Image and Procedure Reuse:** Using a consistent style of imagery allows procedures to be reused across user manuals where applicable. A library of images and procedures was developed for the tech writing team to draw from, reducing development time and building consistency across documents and brands.
3. **Visual Clarity:** I moved away from photography in favor of 3D models. Compared to product photography, the 3D models carry far less visual clutter, and let us reveal internal structure on demand, either by ghosting surrounding parts or using section views.
4. **Accessibility:** For color-blind users, or those who choose to print the manuals in monochrome, all procedures remain easy to follow through the thoughtful use of before/after imagery, iconography, and focused text. These elements reinforce the action - they do not substitute for demonstrating it.